

Environmental Impacts of Spacecraft Launches and Re-entries



Launch and Re-entry Pollutants and Impacts

Launches (0 km to orbit)



Hydrogen
Kerosene
Methane
Hypergolic
Solid
Bio-propellants

H_2O
 Al_2O_3

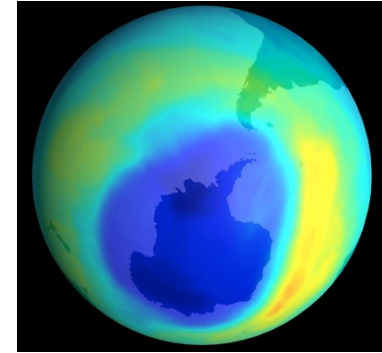
Re-entries (60-80 km)

Payloads
Components
Capsules
Rocket Bodies
Debris



BC
 NO_x
Chlorine
Metal oxides

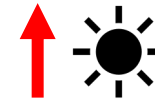
Stratospheric Ozone Loss



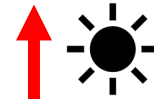
Affected by all emitted compounds

Instantaneous Climate Forcing

BC



Al_2O_3



Stratospheric
clouds



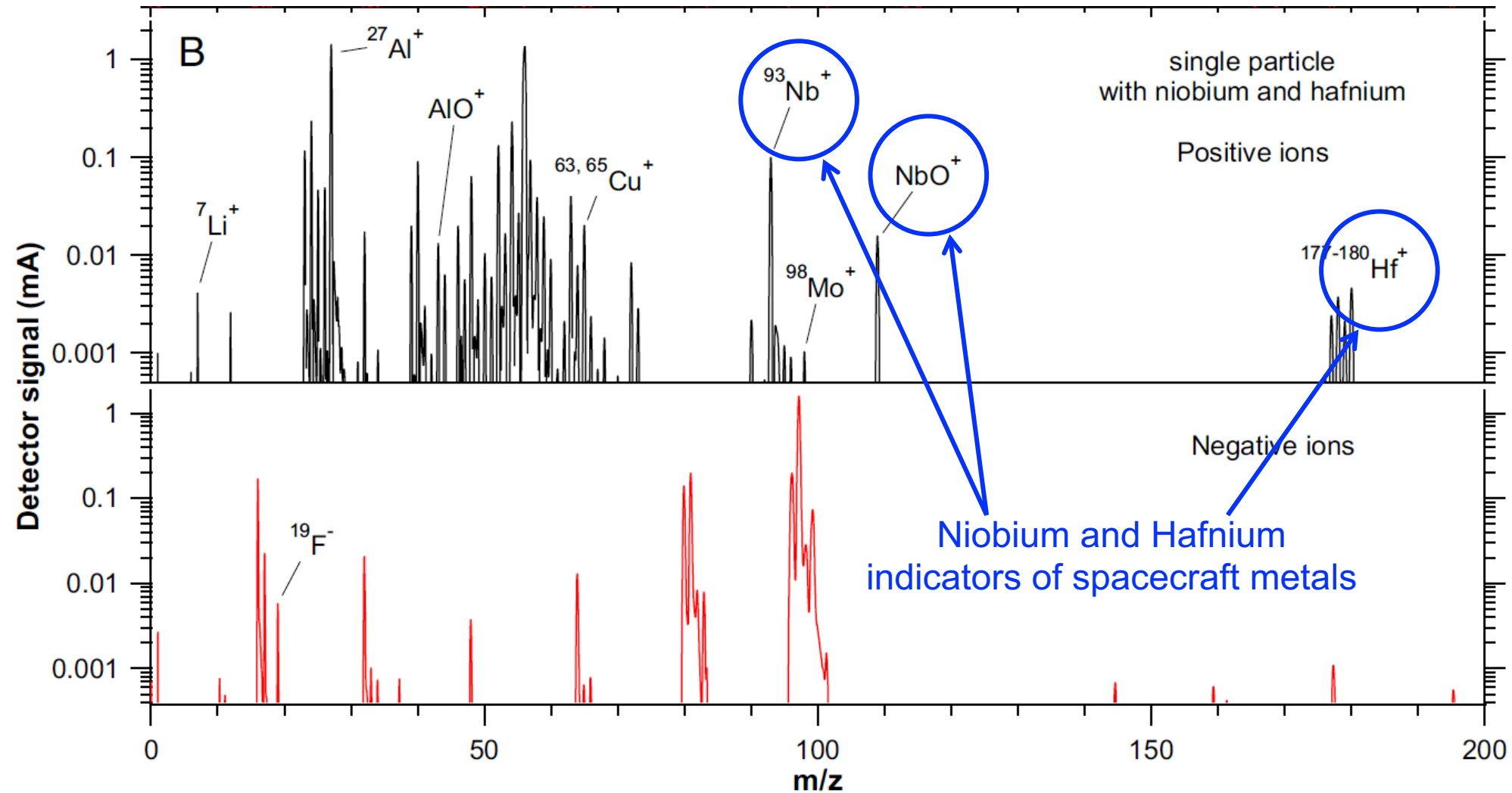
Al_2O_3 : alumina;

BC: black carbon (soot);

NO_x : Nitrogen Oxides ($NO + NO_2$)

Observational Evidence of Contamination from Re-entries

SABRE campaign aircraft measurements of aerosol composition as evidence of influence

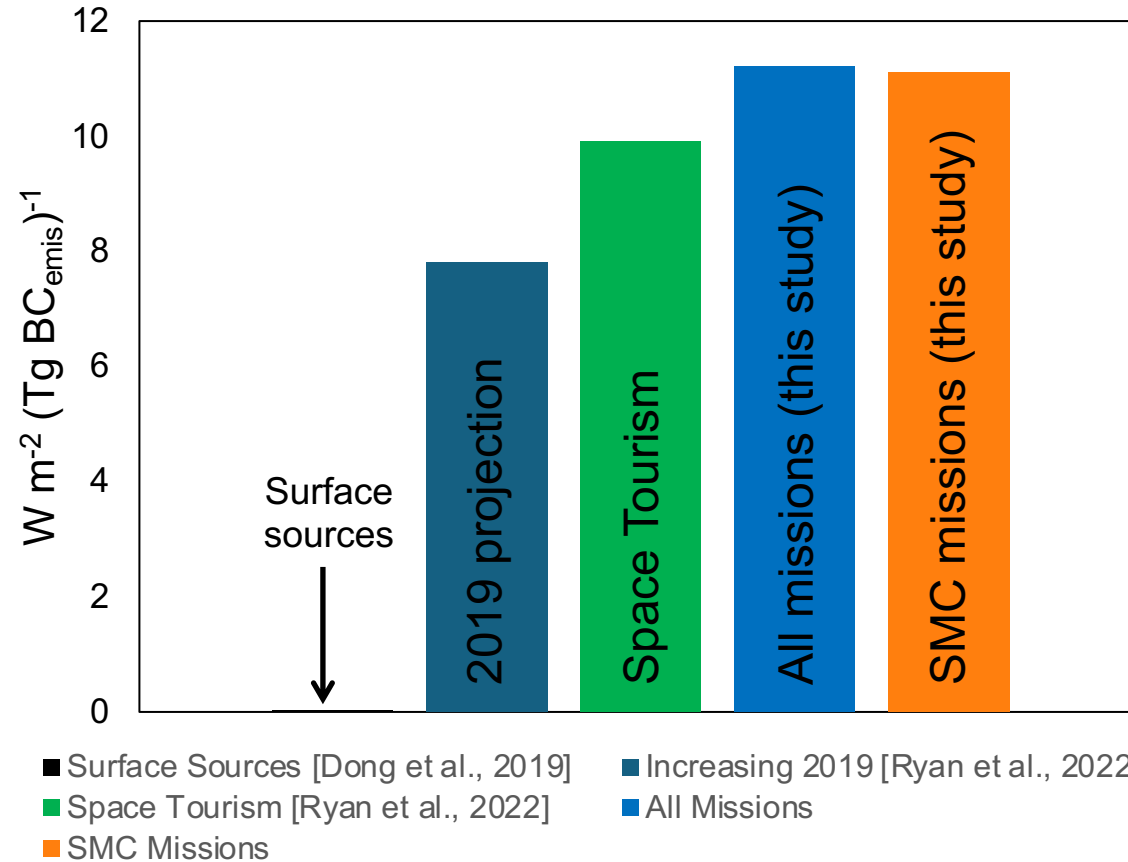


10% of stratospheric aerosols sampled contaminated with metals from spacecraft re-entry pollution

[Murphy et al., 2023]

Contextualizing Launch Emissions Climate Impact

Climate impact of black carbon (soot) particles from Earth-bound sources and rockets

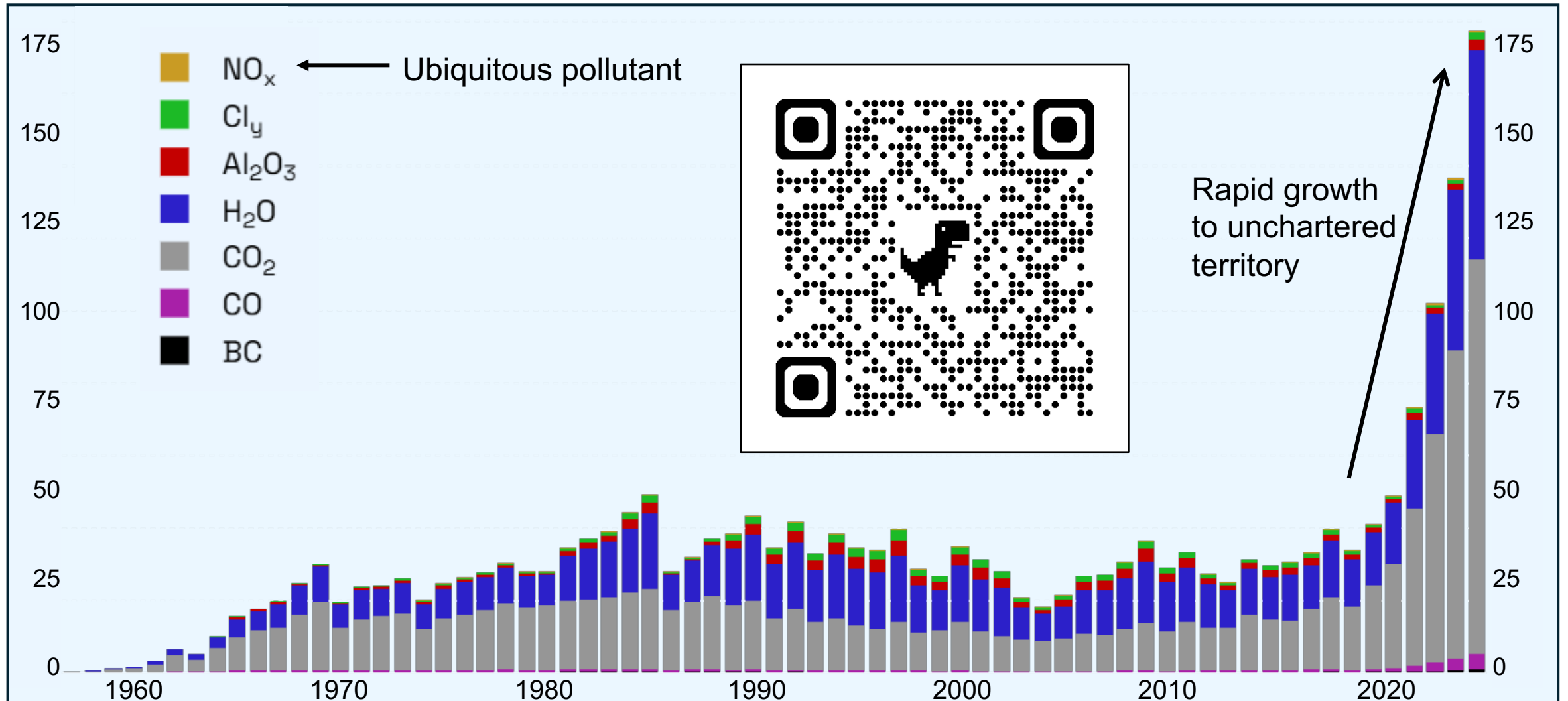


Climate impact of black carbon from rockets **~540 times greater** than all Earth-bound sources

Implications for space sector growth!

Growth in Pollutant Emissions from Launches

Rocket launch air pollutant and CO₂ emissions in kilotonnes from 1957 to 2025



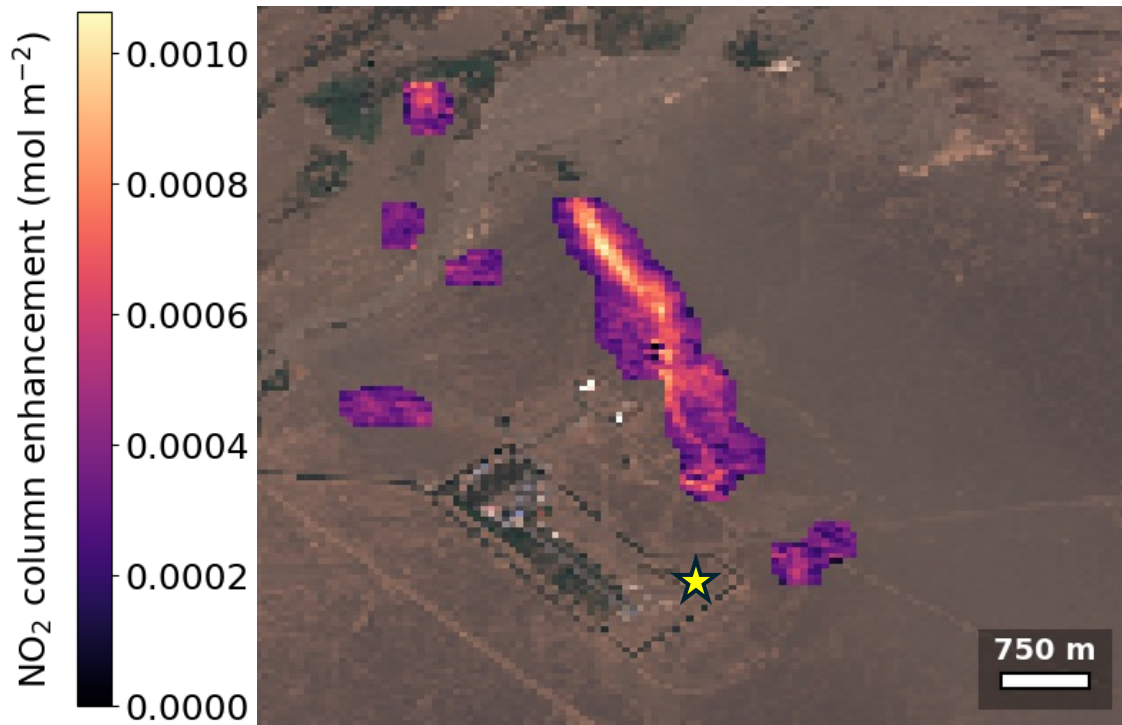
[<https://orbitalemissions.org/launches.html>]

Emissions data are uncertain, outdated, under-validated. Urgent need for observational constraints!

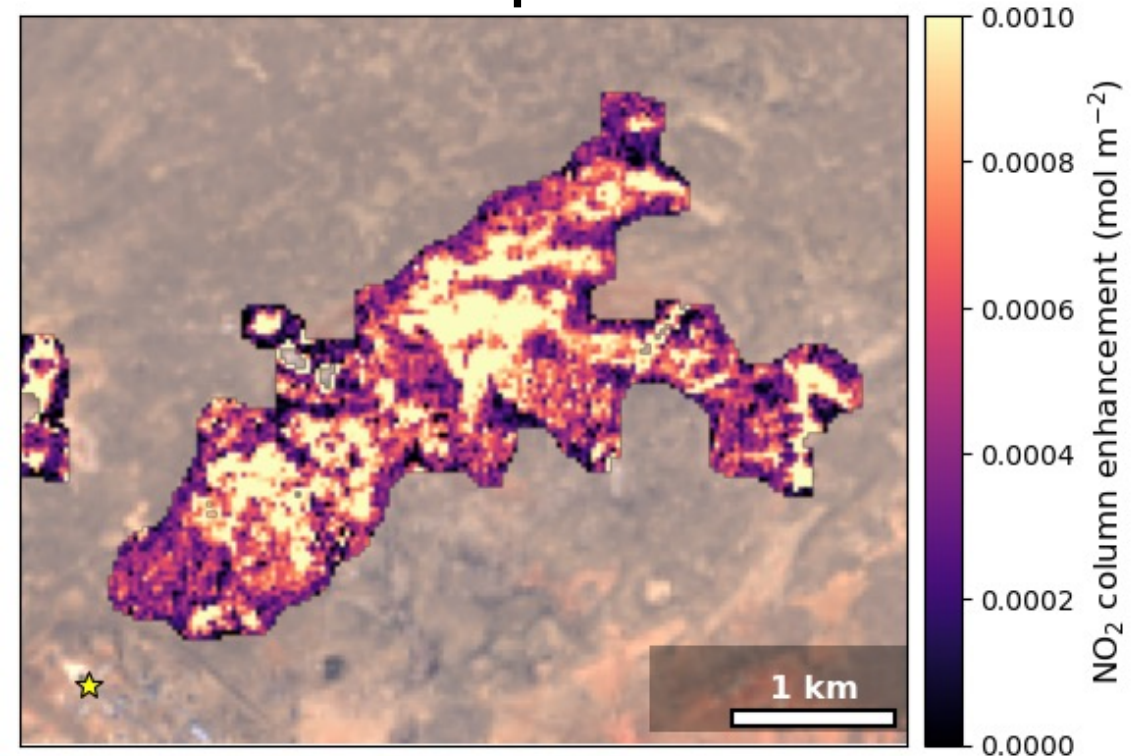
Launch Pollution Observed with Space-based Sensors

Retrieve concentrations of nitrogen dioxide (NO_2) and use to infer nitrogen oxides (NO_x) emissions

China launch pad:



Kazakhstan launch pad:



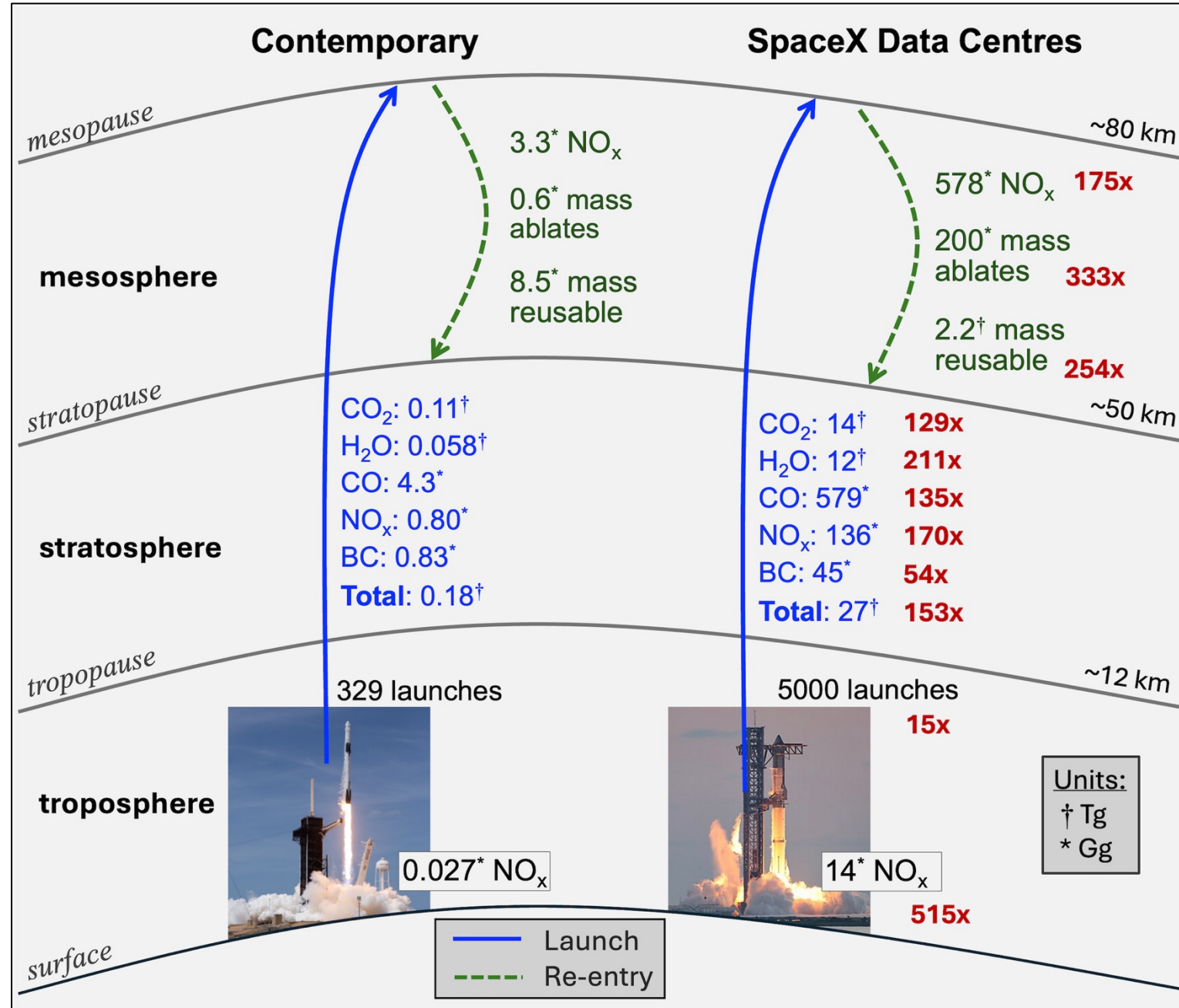
Satellite-derived emissions: 448 (355-550) kg
Inventory emissions: 1450 kg
Propellant burned: 183 tonnes

Satellite-derived emissions: 441 (350-542) kg
Inventory emissions: 1438 kg
Propellant burned: 162 tonnes

Plume resolved to <100 m

What if 1 million data centres launched to orbit?

LEFT: Current emissions



RIGHT: Future emissions

Key Take-Aways

Absolute climate harm is currently small

Black carbon climate impact from rockets at least 500-times more than Earth-bound sources

Re-entry ablation metals contaminate 10% of stratospheric particles

Observations of atmospheric harm are sparse

Rely mostly on models and innovative use of existing observations

Research needs more funding to address knowledge gaps and keep pace with technology

Regulation needs to advance to include environmental impact assessments

Details of Cited Sources

Ryan et al., Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and global climate, Earth's Future, 10, e2021EF002612, doi:10.1029/2021EF002612, 2022.

Barker et al., Global 3D rocket launch and re-entry air pollutant and CO2 emissions at the onset of the megaconstellation era, Nature Scientific Data, 11, 1-15, doi:10.1038/s41597-024-03910-z, 2024.

Barker et al., Radiative forcing and ozone depletion of a decade of satellite megaconstellation missions, Earth's Future, 14, doi:10.1029/2025EF007229, 2026.

Emissions Tracker: <https://orbitalemissions.org/>

Murphy et al., Metals from spacecraft reentry in stratospheric aerosol particles, PNAS, 120(43), doi:10.1073/pnas.2313374120, 2023.

ARIA funded project details here: https://maraisresearchgroup.co.uk/rocket_emis.html and here: <https://aria.org.uk/opportunity-spaces/scoping-our-planet/opportunity-seeds?cardId=unlocking-potential-to-quantify-spacecraft-pollution>